```
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ cat $HADOOP_HOME/etc/hadoop/workers
localhost
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [VenkataPramod]
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ jps
5508 DataNode
5958 Jps
5751 SecondaryNameNode
5375 NameNode
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ hdfs dfs -ls /
Found 4 items
drwxr-xr-x   - pramo supergroup          0 2026-09-10 01:08 /finsight
drwxr-xr-x   - pramo supergroup          0 2026-08-31 10:46 /home
drwxr-xr-x   - pramo supergroup          0 2026-08-31 02:25 /logs
drwxr-xr-x   - pramo supergroup          0 2026-08-31 02:25 /topics
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ hdfs dfs -mkdir -p /user/clouduser/ecommerce/raw/orders
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ hdfs dfs -mkdir -p /user/clouduser/ecommerce/raw/order_items
hdfs dfs -mkdir -p /user/clouduser/ecommerce/raw/payments
hdfs dfs -mkdir -p /user/clouduser/ecommerce/raw/customers
hdfs dfs -mkdir -p /user/clouduser/ecommerce/processed

pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ hdfs dfs -ls -R /user/clouduser/ecommerce
drwxr-xr-x   - pramo supergroup          0 2026-09-15 09:56 /user/clouduser/ecommerce/processed
drwxr-xr-x   - pramo supergroup          0 2026-09-15 09:56 /user/clouduser/ecommerce/raw
drwxr-xr-x   - pramo supergroup          0 2026-09-15 09:56 /user/clouduser/ecommerce/raw/customers
drwxr-xr-x   - pramo supergroup          0 2026-09-15 09:56 /user/clouduser/ecommerce/raw/order_items
drwxr-xr-x   - pramo supergroup          0 2026-09-15 09:56 /user/clouduser/ecommerce/raw/orders
drwxr-xr-x   - pramo supergroup          0 2026-09-15 09:56 /user/clouduser/ecommerce/raw/payments
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$
```

```
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ hdfs dfs -D dfs.blocksize=268435456 -D dfs.replication=1 -put
orders.csv /user/clouduser/ecommerce/raw/orders/
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ hdfs dfs -ls /user/clouduser/ecommerce/raw/orders/
Found 1 items
-rw-r--r--   1 pramo supergroup    9518611 2026-09-15 10:00 /user/clouduser/ecommerce/raw/orders/orders.csv
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ hdfs fsck /user/clouduser/ecommerce/raw/orders/orders.csv -fil
es -blocks -locations
Connecting to namenode via http://localhost:9870/fsck?ugi=pramo&files=1&blocks=1&locations=1&path=%2Fuser%2Fclouduser%2Fecommerce%2Fraw%2Forders%2Forders.cs
v
FSCK started by pramo (auth:SIMPLE) from /127.0.0.1 for path /user/clouduser/ecommerce/raw/orders/orders.csv at Tue Sep 15 10:00:52 IST 2026

/user/clouduser/ecommerce/raw/orders/orders.csv 9518611 bytes, replicated: replication=1, 1 block(s):  OK
0. BP-160395664-127.0.1.1-1788084690157:blk_1073747958_7135 len=9518611 Live_repl=1  [DatanodeInfoWithStorage[127.0.0.1:9866,DS-f41e2aa7-04ab-471e-8087-7dd0
81b13acd,DISK]]


Status: HEALTHY
 Number of data-nodes:  1
 Number of racks:               1
 Total dirs:                    0
 Total symlinks:                0

Replicated Blocks:
 Total size:    9518611 B
 Total files:   1
 Total blocks (validated):      1 (avg. block size 9518611 B)
 Minimally replicated blocks:   1 (100.0 %)
 Over-replicated blocks:        0 (0.0 %)
 Under-replicated blocks:       0 (0.0 %)
 Mis-replicated blocks:         0 (0.0 %)
 Default replication factor:    1
 Average block replication:     1.0
 Missing blocks:                0
 Corrupt blocks:                0
 Missing replicas:              0 (0.0 %)
 Blocks queued for replication: 0
```

```
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ hdfs dfs -put customers.csv /user/clouduser/ecommerce/raw/cust
omers/
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ hdfs dfs -ls /user/clouduser/ecommerce/raw/customers/
Found 1 items
-rw-r--r--   1 pramo supergroup    2953236 2026-09-15 10:03 /user/clouduser/ecommerce/raw/customers/customers.csv
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ hdfs dfs -du -h /user/clouduser/ecommerce
0       0       /user/clouduser/ecommerce/processed
11.9 M  11.9 M  /user/clouduser/ecommerce/raw
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ hdfs dfs -cp /user/clouduser/ecommerce/raw/orders/orders.csv /
user/clouduser/ecommerce/processed/
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ hdfs dfs -ls /user/clouduser/ecommerce/raw/orders/
Found 1 items
-rw-r--r--   1 pramo supergroup    9518611 2026-09-15 10:00 /user/clouduser/ecommerce/raw/orders/orders.csv
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ hdfs dfs -ls /user/clouduser/ecommerce/processed/
Found 1 items
-rw-r--r--   1 pramo supergroup    9518611 2026-09-15 10:05 /user/clouduser/ecommerce/processed/orders.csv
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$
```

```
pramo@VenkataPramod:/mnt/c/Users/pramo$ hdfs dfs -ls /user/clouduser/ecommerce/processed/city_payment_summary
Found 2 items
-rw-r--r--   1 pramo supergroup          0 2026-09-15 11:39 /user/clouduser/ecommerce/processed/city_payment_summary/_SUCCESS
-rw-r--r--   1 pramo supergroup        648 2026-09-15 11:39 /user/clouduser/ecommerce/processed/city_payment_summary/part-00000-c94df39c-02f8-4fc9-9afd-3da2
d91d93f1-c000.snappy.parquet
pramo@VenkataPramod:/mnt/c/Users/pramo$
```

```
pramo@VenkataPramod:/mnt/c/Users/pramo$ hdfs dfs -ls /user/clouduser/ecommerce/processed/delivery_analysis
Found 2 items
-rw-r--r--   1 pramo supergroup          0 2026-09-15 11:42 /user/clouduser/ecommerce/processed/delivery_analysis/_SUCCE
SS
-rw-r--r--   1 pramo supergroup       2150 2026-09-15 11:42 /user/clouduser/ecommerce/processed/delivery_analysis/part-0
0000-bbd34af6-9d2f-4036-8ce2-aa2887d5a399-c000.snappy.parquet
pramo@VenkataPramod:/mnt/c/Users/pramo$
```

```
pramo@VenkataPramod:~/kafka$ jps
5508 DataNode
10231 SparkSubmit
5751 SecondaryNameNode
16920 Kafka
16955 Jps
5375 NameNode
pramo@VenkataPramod:~/kafka$ kafka-topics.sh --create --topic order-stream --bootstrap-server localhost
:9092 --partitions 1 --replication-factor 1
Created topic order-stream.
pramo@VenkataPramod:~/kafka$ kafka-topics.sh --list --bootstrap-server localhost:9092
order-stream
pramo@VenkataPramod:~/kafka$
```

```
pramo@VenkataPramod:~/kafka$ cd "/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata"
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ ls -lh orders.csv
-rwxrwxrwx 1 pramo pramo 9.1M Sep 15 09:22 orders.csv
pramo@VenkataPramod:/mnt/c/Users/pramo/Downloads/Bigdata Haddop Spark/Input files & metadata$ kafka-console-producer.sh --topic order-str
eam --bootstrap-server localhost:9092
```

```
er, 10.255.255.254, 37665, None)
26/09/15 11:55:13 INFO BlockManagerMaster: Registere
26/09/15 11:55:13 INFO BlockManager: Initialized Blo
26/09/15 11:55:19 WARN ResolveWriteToStream: spark.s
abled.
-------------------------------------------
Batch: 0
-------------------------------------------
+------------+-----------+
|order_status|order_count|
+------------+-----------+
|shipped     |936        |
|canceled    |409        |
|approved    |2          |
|invoiced    |266        |
|delivered   |87428      |
|unavailable |2          |
|processing  |273        |
+------------+-----------+
```

```
ecomdb_pramodvchalla
Time taken: 0.974 seconds, Fetched 2 row(s)
spark-sql (default)> USE ecomdb_pramodvchalla;
Time taken: 0.057 seconds
spark-sql (ecomdb_pramodvchalla)> CREATE TABLE orders_raw (
                                >     order_id STRING,
                                >     customer_id STRING,
                                >     order_status STRING,
                                >     order_purchase_timestamp STRING,
                                >     order_approved_at STRING,
                                >     order_delivered_timestamp STRING,
                                >     order_estimated_delivery_date STRING
                                > )
                                > USING csv
                                > OPTIONS (
                                >     path 'hdfs:///user/clouduser/ecommerce/raw/orders/',
                                >     header 'true',
                                >     inferSchema 'false'
                                > );
Time taken: 0.737 seconds
spark-sql (ecomdb_pramodvchalla)> show tables;
orders_raw
Time taken: 0.186 seconds, Fetched 1 row(s)
spark-sql (ecomdb_pramodvchalla)> SELECT * FROM orders_raw LIMIT 5;
Axfy13Hk4PIk    hCT0x9JiGXBQ    delivered       2017-10-22 18:57:54     2017-10-22 19:14:13     2017-10-26 22:19:52     2017-11-09
v6px92oS8cLG    PxA7fv9spyhx    delivered       2018-06-20 21:40:31     2018-06-20 22:20:20     2018-07-03 22:51:22     2018-07-24
Ulpf9skrhjfm    g3nXeJkGI0Qw    delivered       2018-02-16 16:19:31     2018-02-17 16:15:35     2018-02-27 01:29:50     2018-03-08
bwJVWupf2keN    EOEsCQ6QlpIg    delivered       2018-08-18 18:04:29     2018-08-18 18:15:16     2018-08-27 20:03:51     2018-09-19
Dd0QnrMk9Cj5    mVz5LO2Vd6cL    delivered       2017-12-22 16:44:04     2017-12-22 17:31:31     2018-01-05 19:22:49     2018-01-18
Time taken: 2.26 seconds, Fetched 5 row(s)
spark-sql (ecomdb_pramodvchalla)>
```

File Edit View Options Add-Ons Help

Search for tools, help, and resources...

Favorites | In/Out | Preparation | Join | Parse | Transform | In-Database | Reporting | Documentation | Spatial | AI | Interface | Data Investigation | Connectors | Address | Demographic Analysis | Calgary

Auto Insights Uploader | Browse | Date Time Now | Directory | Input Data | Map Input | Output Data | Text Input

Output Data (9) - Configuration

Write to File or Database

C:\Users\pramo\Downloads\Bigdata Haddop Spark\Input files & metadata\

Save to a File | Connect to Data

Options

	Name	Value
1	Max Records Per File	
2	File Format	Comma Separated Value (*.csv
3	Delimiters	,
4	First Row Contains Field Names	☑
5	Quote Output Fields	Auto
6	Code Page	ISO 8859-1 Latin I
7	Line Ending Style	Windows
8	Write BOM	☑
9	Suppress Output if No Records	☐

☐ Take File/Table Name From Field

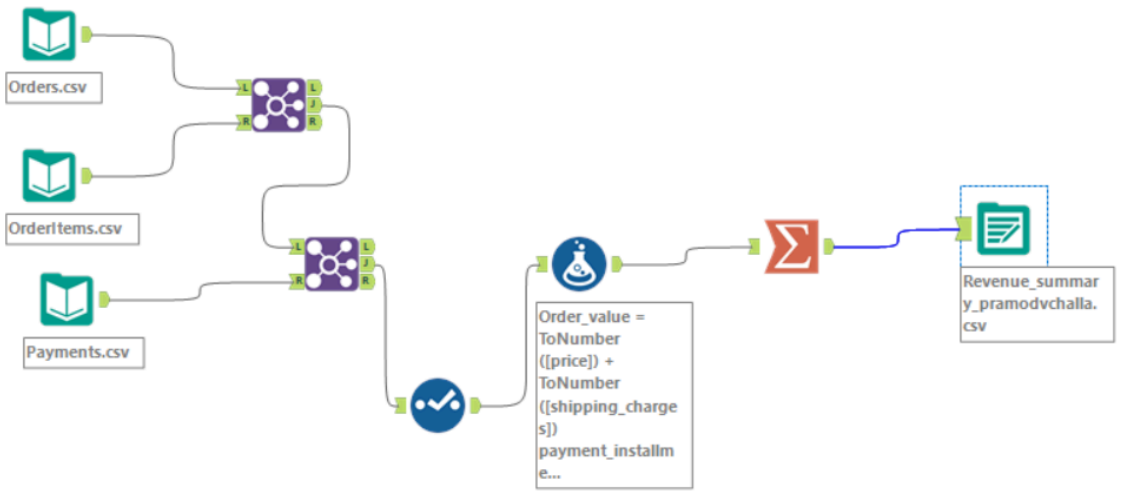
Append Suffix to File/Table Name

Field Containing File Name or Part of File Name

Sum_Order_value ☑ Keep Field in Output

ew Workflow2 × | New Workflow3 × | New Workflow4 × | New Workflow5 × | New Workflow6 × | *WF1_pramodvchalla.yxmd × | *WF2_pramodvchalla.yxmd ×

Run



Results - Output Data (9) - Input

3 of 3 Fields | Cell Viewer | 1 record displayed | Search | Data | Metadata | Actions

Record	Sum_Order_value	Count	AvgNo0_payment_installments
1	34403072.06	89316	2.965817

File Edit View Options Add-Ons Help

Search for tools, help, and resources...

Favorites | In/Out | Preparation | Join | Parse | Transform | In-Database | Reporting | Documentation | Spatial | AI | Interface | Data Investigation | Connectors | Address | Demographic Analysis | Calgary

Auto Insights Uploader | Browse | Date Time Now | Directory | Input Data | Map Input | Output Data | Text Input

Workflow - Configuration

Meta Info

Canvas | Workflow | Runtime | Events

CANVAS OPTIONS

Layout Direction

Horizontal

Annotations

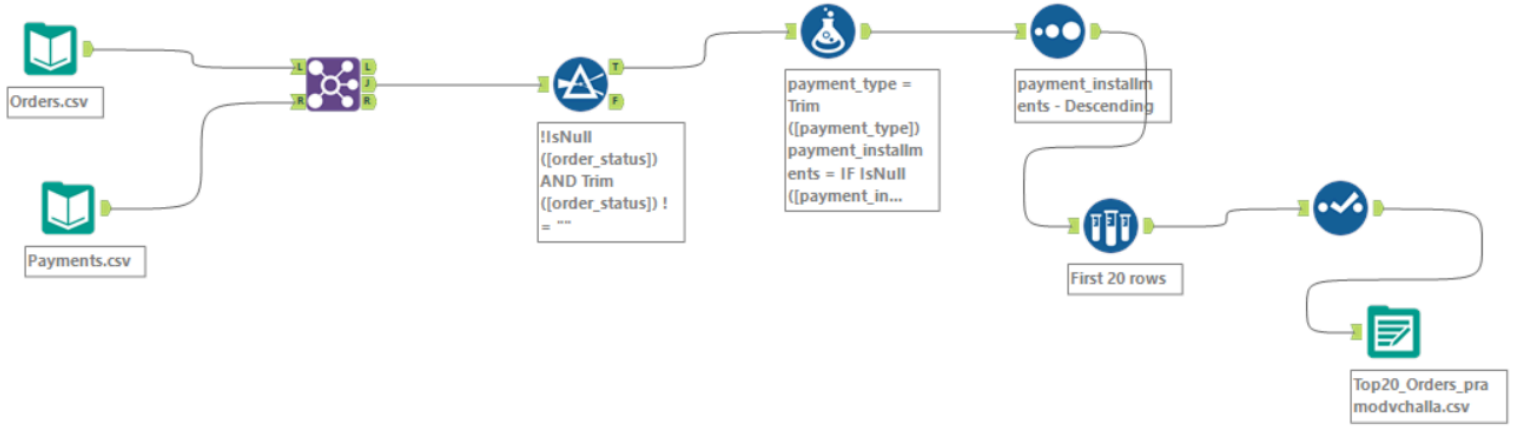
Show

Connection Progress

Show Only When Running

New Workflow1 | New Workflow2 | New Workflow3 | New Workflow4 | New Workflow5 | New Workflow6 | *WF1_pramodvchalla.yxmd

Run



```
spark-sql (ecomdb_pramodvchalla)> INSERT INTO orders_part
                                > SELECT
                                >     order_id,
                                >     customer_id,
                                >     order_status,
                                >     order_purchase_timestamp,
                                >     order_approved_at,
                                >     order_delivered_timestamp,
                                >     order_estimated_delivery_date,
                                >     YEAR(TO_TIMESTAMP(order_purchase_timestamp)),
                                >     MONTH(TO_TIMESTAMP(order_purchase_timestamp))
                                > FROM orders_raw;
Time taken: 11.171 seconds
spark-sql (ecomdb_pramodvchalla)> SHOW PARTITIONS orders_part;
order_year=2016/order_month=10
order_year=2016/order_month=12
order_year=2016/order_month=9
order_year=2017/order_month=1
order_year=2017/order_month=10
order_year=2017/order_month=11
order_year=2017/order_month=12
order_year=2017/order_month=2
order_year=2017/order_month=3
order_year=2017/order_month=4
order_year=2017/order_month=5
order_year=2017/order_month=6
order_year=2017/order_month=7
order_year=2017/order_month=8
order_year=2017/order_month=9
order_year=2018/order_month=1
order_year=2018/order_month=2
order_year=2018/order_month=3
order_year=2018/order_month=4
order_year=2018/order_month=5
order_year=2018/order_month=6
order_year=2018/order_month=7
order_year=2018/order_month=8
order_year=2018/order_month=9
Time taken: 0.036 seconds, Fetched 24 row(s)
spark-sql (ecomdb_pramodvchalla)>
```

```
spark-sql (ecomdb_pramodvchalla)> SELECT
                                >     order_status,
                                >     COUNT(*) AS order_count
                                > FROM orders_part
                                > GROUP BY order_status
                                > ORDER BY order_count DESC;
delivered       87428
shipped 936
canceled        409
processing      273
invoiced        266
approved        2
unavailable     2
Time taken: 4.005 seconds, Fetched 7 row(s)
spark-sql (ecomdb_pramodvchalla)> CREATE TABLE customer (
                                >     customer_id STRING,
                                >     customer_zip_code_prefix STRING,
                                >     customer_city STRING,
                                >     customer_state STRING
                                > )
                                > USING csv
                                > OPTIONS (
                                >     path 'hdfs:///user/clouduser/ecommerce/raw/customers/',
                                >     header 'true',
                                >     inferSchema 'false'
                                > );
Time taken: 0.052 seconds
spark-sql (ecomdb_pramodvchalla)> SHOW TABLES;
customer
orders_part
orders_raw
Time taken: 0.03 seconds, Fetched 3 row(s)
spark-sql (ecomdb_pramodvchalla)> SELECT * FROM customer LIMIT 5;
hCT0x9JiGXBQ    58125   varzea paulista SP
PxA7fv9spyhx    3112    armacao dos buzios      RJ
g3nXeJkGI0Qw    4119    jandira SP
EOEsCQ6QlpIg    18212   uberlandia      MG
mVz5LO2Vd6cL    88868   ilhabela        SP
Time taken: 0.118 seconds, Fetched 5 row(s)
spark-sql (ecomdb_pramodvchalla)>
```

```
spark-sql (ecomdb_pramodvchalla)> SHOW TABLES;
customer
orders_part
orders_raw
Time taken: 0.03 seconds, Fetched 3 row(s)
spark-sql (ecomdb_pramodvchalla)> SELECT * FROM customer LIMIT 5;
hCT0x9JiGXBQ    58125   varzea paulista SP
PxA7fv9spyhx    3112    armacao dos buzios      RJ
g3nXeJkGI0Qw    4119    jandira SP
EOEsCQ6QlpIg    18212   uberlandia      MG
mVz5LO2Vd6cL    88868   ilhabela        SP
Time taken: 0.118 seconds, Fetched 5 row(s)
spark-sql (ecomdb_pramodvchalla)> SELECT
                                >     c.customer_city,
                                >     COUNT(DISTINCT o.order_id) AS delivered_unique_orders
                                > FROM customer c
                                > JOIN orders_part o
                                >     ON c.customer_id = o.customer_id
                                > WHERE LOWER(o.order_status) = 'delivered'
                                > GROUP BY c.customer_city
                                > ORDER BY delivered_unique_orders DESC
                                > LIMIT 10;
sao paulo       14020
rio de janeiro  6062
belo horizonte  2380
brasilia        1740
curitiba        1405
porto alegre    1290
campinas        1283
salvador        1115
guarulhos       1045
sao bernardo do campo   829
Time taken: 3.198 seconds, Fetched 10 row(s)
spark-sql (ecomdb_pramodvchalla)>
```